FA24-BCS-070

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Lab 2 — Responsive Navbar and Grid Drill

Difficulty: Easy–Medium · **Covers:** Lectures 7–10 · **Folder:** lab-2/

A focused, smaller-scale rehearsal for Assignment 2: build **just** a responsive navbar (logo left, links center, cart icon right, collapsing into a hamburger menu below 768px) and a 4-column-desktop / 2-column-tablet / 1-column-phone product grid, using Flexbox or Grid plus Bootstrap or Tailwind. Content can be placeholder text/images — the point is the responsive mechanics, not the final design.

Deliverables in lab-2/: index.html, your CSS/framework files, and your completion PDF (with screenshots at phone/tablet/desktop widths).

CODE:

index.html:

```
<!doctype html>
<html lang="en">
 <head>
  <meta charset="UTF-8" />
  <meta
   name="viewport"
   content="width=device-width, initial-scale=1.0, maximum-scale=1.0,
user-scalable=no"
  />
  <title>Lab 2 - Responsive Navbar & Grid</title>
  <link rel="stylesheet" href="css/bootstrap.min.css" />
  <link rel="stylesheet" href="css/bootstrap-icons.min.css" />
  <link rel="stylesheet" href="css/styles.css" />
 </head>
 <body>
  <header>
   <nav class="navbar apple-nav fixed-top border-bottom">
    <div class="container nav-shell">
     <a class="navbar-brand text-dark me-3" href="#" aria-label="Home">
      <i class="bi bi-apple fs-5"></i>
     </a>
```

```
<input type="checkbox" id="nav-toggle" class="nav-toggle" />
<label
 for="nav-toggle"
 class="nav-toggle-label"
 aria-label="Toggle navigation"
>
 <span></span>
 <span></span>
 <span></span>
</label>

<div class="nav-menu">
 <ul class="navbar-nav gap-2 gap-md-3 flex-row">
  <li class="nav-item">
   <a class="nav-link text-secondary px-2 small" href="#">Store</a>
  </li>
  <li class="nav-item">
   <a class="nav-link text-secondary px-2 small" href="#products"
    >Mac</a
   >
  </li>
  <li class="nav-item">
   <a class="nav-link text-secondary px-2 small" href="#products"
    >iPad</a
   >
```

```
    </li>
    <li class="nav-item">
      <a class="nav-link text-secondary px-2 small" href="#products"
        >iPhone</a
      >
    </li>
    <li class="nav-item">
      <a class="nav-link text-secondary px-2 small" href="#products"
        >Watch</a
      >
    </li>
    <li class="nav-item">
      <a class="nav-link text-secondary px-2 small" href="#products"
        >AirPods</a
      >
    </li>
    <li class="nav-item">
      <a class="nav-link text-secondary px-2 small" href="#products"
        >Accessories</a
      >
    </li>
  </ul>
</div>

<div class="d-flex align-items-center cart-wrap">
```

```
          <a
            href="#"
            class="text-dark position-relative text-decoration-none"
            aria-label="Shopping bag"
          >
            <i class="bi bi-bag fs-5"></i>
            <span
              class="position-absolute top-0 start-100 translate-middle badge rounded-pill bg-dark"
              style="font-size: 0.6rem"
              >0</span
            >
          </a>
        </div>
      </div>
    </nav>
  </header>

  <main>
    <section class="hero bg-body-tertiary py-5 text-center">
      <div class="container py-3">
        <h1 class="display-5 fw-bold">
          Store.
          <span class="text-secondary fw-normal"
            >The best way to buy the products you love.</span
```

```
        >
       </h1>
       <p class="text-secondary fs-5 mt-2 mb-0">
        Engineered with Apple silicon. Exceptional performance in every
        device.
       </p>
      </div>
     </section>

     <section class="py-5" id="products">
      <div class="container">
       <div class="d-flex justify-content-between align-items-baseline mb-
4">
        <h2 class="h3 fw-bold m-0">All Products</h2>
        <span class="text-secondary small">8 models available</span>
       </div>

       <div class="row row-cols-1 row-cols-md-2 row-cols-lg-4 g-4">
        <div class="col">
         <article
          class="card h-100 border rounded-4 overflow-hidden product-card"
         >
          <div
           class="bg-body-tertiary p-3 d-flex align-items-center justify-
content-center card-image-box"
          >
```

```
<img
 src="https://images.unsplash.com/photo-1517336714731-
489689fd1ca8?auto=format&fit=crop&w=600&q=80"
 class="img-fluid object-fit-cover rounded-3 mh-100"
 alt="MacBook Pro 16-inch"
/>
</div>
<div class="card-body d-flex flex-column p-4">
<span
 class="text-uppercase fw-bold text-danger small mb-1"
 style="font-size: 0.72rem"
 >New M3 Max</span
>
<h3 class="h5 fw-bold mb-1">
 <a href="#" class="text-decoration-none text-dark"
  >MacBook Pro 16"</a
 >
</h3>
<p class="text-secondary small mb-3 flex-grow-1">
 Mind-blowing performance with Liquid Retina XDR display.
</p>
<div class="fw-bold mb-3">From $2,499</div>
<button
 type="button"
 class="btn btn-apple w-100 py-2 fw-medium border-0"
```

```
          >
            Add to Bag
          </button>
        </div>
      </article>
    </div>

    <div class="col">
      <article
        class="card h-100 border rounded-4 overflow-hidden product-card"
      >
        <div
          class="bg-body-tertiary p-3 d-flex align-items-center justify-content-center card-image-box"
        >
          <img
            src="https://images.unsplash.com/photo-1611186871348-b1ce696e52c9?auto=format&fit=crop&w=600&q=80"
            class="img-fluid object-fit-cover rounded-3 mh-100"
            alt="MacBook Air 15-inch"
          />
        </div>
        <div class="card-body d-flex flex-column p-4">
          <span
            class="text-uppercase fw-bold text-danger small mb-1"
            style="font-size: 0.72rem"
```

```
      >Thin &amp;amp; Powerful</span
    >
    <h3 class="h5 fw-bold mb-1">
     <a href="#" class="text-decoration-none text-dark"
      >MacBook Air 15"</a
     >
    </h3>
    <p class="text-secondary small mb-3 flex-grow-1">
     Incredibly thin design and up to 18 hours of battery life.
    </p>
    <div class="fw-bold mb-3">From $1,299</div>
    <button
     type="button"
     class="btn btn-apple w-100 py-2 fw-medium border-0"
    >
     Add to Bag
    </button>
   </div>
  </article>
</div>

<div class="col">
 <article
  class="card h-100 border rounded-4 overflow-hidden product-card"
 >
```

```
<div
 class="bg-body-tertiary p-3 d-flex align-items-center justify-content-center card-image-box"
>
 <img
  src="https://images.unsplash.com/photo-1592750475338-74b7b21085ab?auto=format&fit=crop&w=600&q=80"
  class="img-fluid object-fit-cover rounded-3 mh-100"
  alt="iPad Pro"
 />
</div>
<div class="card-body d-flex flex-column p-4">
 <span
  class="text-uppercase fw-bold text-danger small mb-1"
  style="font-size: 0.72rem"
  >Ultra Retina</span
 >
 <h3 class="h5 fw-bold mb-1">
  <a href="#" class="text-decoration-none text-dark"
   >iPad Pro</a
  >
 </h3>
 <p class="text-secondary small mb-3 flex-grow-1">
  A stunningly advanced tablet for work and creativity.
 </p>
 <div class="fw-bold mb-3">From $999</div>
```

```
            <button
              type="button"
              class="btn btn-apple w-100 py-2 fw-medium border-0"
            >
              Add to Bag
            </button>
          </div>
        </article>
      </div>

      <div class="col">
        <article
          class="card h-100 border rounded-4 overflow-hidden product-card"
        >
          <div
            class="bg-body-tertiary p-3 d-flex align-items-center justify-content-center card-image-box"
          >
            <img
              src="https://images.unsplash.com/photo-1510557880182-3d4d3cba35a5?auto=format&fit=crop&w=600&q=80"
              class="img-fluid object-fit-cover rounded-3 mh-100"
              alt="iPhone 15 Pro"
            />
          </div>
          <div class="card-body d-flex flex-column p-4">
```

```
    <span
     class="text-uppercase fw-bold text-danger small mb-1"
     style="font-size: 0.72rem"
     >Titanium</span
    >
    <h3 class="h5 fw-bold mb-1">
     <a href="#" class="text-decoration-none text-dark"
      >iPhone 15 Pro</a
     >
    </h3>
    <p class="text-secondary small mb-3 flex-grow-1">
     The most powerful iPhone with pro-level camera features.
    </p>
    <div class="fw-bold mb-3">From $999</div>
    <button
     type="button"
     class="btn btn-apple w-100 py-2 fw-medium border-0"
    >
     Add to Bag
    </button>
   </div>
  </article>
</div>

<div class="col">
```

```
<article
 class="card h-100 border rounded-4 overflow-hidden product-card"
>
 <div
  class="bg-body-tertiary p-3 d-flex align-items-center justify-content-center card-image-box"
 >
  <img
   src="https://images.unsplash.com/photo-1544244015-0df4b3ffc6b0?auto=format&fit=crop&w=600&q=80"
   class="img-fluid object-fit-cover rounded-3 mh-100"
   alt="Apple Watch Ultra"
  />
 </div>
 <div class="card-body d-flex flex-column p-4">
  <span
   class="text-uppercase fw-bold text-danger small mb-1"
   style="font-size: 0.72rem"
   >Adventure</span
  >
  <h3 class="h5 fw-bold mb-1">
   <a href="#" class="text-decoration-none text-dark"
    >Apple Watch Ultra</a
   >
  </h3>
  <p class="text-secondary small mb-3 flex-grow-1">
```

```
            Built for endurance, with precision health tracking.
          </p>
          <div class="fw-bold mb-3">From $799</div>
          <button
           type="button"
           class="btn btn-apple w-100 py-2 fw-medium border-0"
          >
           Add to Bag
          </button>
         </div>
        </article>
       </div>

       <div class="col">
        <article
         class="card h-100 border rounded-4 overflow-hidden product-card"
        >
         <div
          class="bg-body-tertiary p-3 d-flex align-items-center justify-
content-center card-image-box"
          >
          <img
           src="https://images.unsplash.com/photo-1561154464-
82e9adf32764?auto=format&fit=crop&w=600&q=80"
           class="img-fluid object-fit-cover rounded-3 mh-100"
           alt="AirPods Pro"
```

```
  />
 </div>
 <div class="card-body d-flex flex-column p-4">
  <span
   class="text-uppercase fw-bold text-danger small mb-1"
   style="font-size: 0.72rem"
   >Noise Canceling</span
  >
  <h3 class="h5 fw-bold mb-1">
   <a href="#" class="text-decoration-none text-dark"
    >AirPods Pro</a
   >
  </h3>
  <p class="text-secondary small mb-3 flex-grow-1">
   Immersive sound with Adaptive Audio and comfort fit.
  </p>
  <div class="fw-bold mb-3">From $249</div>
  <button
   type="button"
   class="btn btn-apple w-100 py-2 fw-medium border-0"
  >
   Add to Bag
  </button>
 </div>
</article>
```

```
</div>

<div class="col">
 <article
  class="card h-100 border rounded-4 overflow-hidden product-card"
 >
  <div
   class="bg-body-tertiary p-3 d-flex align-items-center justify-content-center card-image-box"
  >
   <img
    src="https://images.unsplash.com/photo-1508685096489-7aacd43bd3b1?auto=format&fit=crop&w=600&q=80"
    class="img-fluid object-fit-cover rounded-3 mh-100"
    alt="Magic Keyboard"
   />
  </div>
  <div class="card-body d-flex flex-column p-4">
   <span
    class="text-uppercase fw-bold text-danger small mb-1"
    style="font-size: 0.72rem"
    >Workspace</span
   >
   <h3 class="h5 fw-bold mb-1">
    <a href="#" class="text-decoration-none text-dark"
     >Magic Keyboard</a
```

```
          >
        </h3>
        <p class="text-secondary small mb-3 flex-grow-1">
          Comfortable typing and seamless connectivity for your setup.
        </p>
        <div class="fw-bold mb-3">From $179</div>
        <button
          type="button"
          class="btn btn-apple w-100 py-2 fw-medium border-0"
        >
          Add to Bag
        </button>
      </div>
    </article>
  </div>

  <div class="col">
    <article
      class="card h-100 border rounded-4 overflow-hidden product-card"
    >
      <div
        class="bg-body-tertiary p-3 d-flex align-items-center justify-content-center card-image-box"
      >
        <img
```

```
          src="https://images.unsplash.com/photo-1600294037681-
c80b4cb5b434?auto=format&fit=crop&w=600&q=80"
          class="img-fluid object-fit-cover rounded-3 mh-100"
          alt="Studio Display"
         />
        </div>
        <div class="card-body d-flex flex-column p-4">
         <span
          class="text-uppercase fw-bold text-danger small mb-1"
          style="font-size: 0.72rem"
          >Display</span
         >
         <h3 class="h5 fw-bold mb-1">
          <a href="#" class="text-decoration-none text-dark"
           >Studio Display</a
          >
         </h3>
         <p class="text-secondary small mb-3 flex-grow-1">
          A brilliant 5K monitor built to elevate every workspace.
         </p>
         <div class="fw-bold mb-3">From $1,599</div>
         <button
          type="button"
          class="btn btn-apple w-100 py-2 fw-medium border-0"
         >
```

```
            Add to Bag
          </button>
        </div>
      </article>
    </div>
  </div>
 </div>
</section>
</main>
</body>
</html>
```

styles.css:

```
:root {
  --apple-dark: #1d1d1f;
  --apple-blue: #0071e3;
  --apple-blue-hover: #0077ed;
  --apple-grey: #f5f5f7;
}

body {
  font-family:
    -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif;
  color: var(--apple-dark);
```

```
  padding-top: 48px;
}

.apple-nav {
  background-color: rgba(255, 255, 255, 0.85);
  backdrop-filter: blur(20px);
  -webkit-backdrop-filter: blur(20px);
  height: 48px;
}

.nav-shell {
  display: flex;
  align-items: center;
  justify-content: space-between;
  flex-wrap: wrap;
}

.navbar-brand {
  z-index: 5;
}

.nav-toggle {
  display: none;
}
```

```
.nav-toggle-label {
  display: none;
  width: 2.5rem;
  height: 2.5rem;
  border-radius: 50%;
  align-items: center;
  justify-content: center;
  flex-direction: column;
  gap: 0.25rem;
  cursor: pointer;
  z-index: 5;
}

.nav-toggle-label span {
  display: block;
  width: 1.1rem;
  height: 2px;
  background-color: var(--apple-dark);
  border-radius: 2px;
}

.nav-menu {
  display: block;
  flex: 1;
}
```

```
.nav-menu .navbar-nav {
  justify-content: center;
  align-items: center;
}

.cart-wrap {
  margin-left: auto;
}

.product-card {
  transition:
    transform 0.25s ease,
    box-shadow 0.25s ease;
}

.product-card:hover {
  transform: translateY(-4px);
  box-shadow: 0 12px 24px rgba(0, 0, 0, 0.08);
}

.card-image-box {
  height: 230px;
}
```

```
.btn-apple {
  background-color: var(--apple-blue);
  color: #ffffff;
  border-radius: 999px;
}

.btn-apple:hover {
  background-color: var(--apple-blue-hover);
  color: #ffffff;
}

@media (max-width: 768px) {
  .apple-nav {
    height: auto;
    min-height: 48px;
  }

  .nav-shell {
    row-gap: 0.5rem;
  }

  .nav-toggle-label {
    display: flex;
    order: 2;
    margin-left: auto;
```

```
}

.cart-wrap {
  order: 3;
}

.nav-menu {
  display: none;
  width: 100%;
  order: 4;
  flex-basis: 100%;
}

.nav-toggle:checked + .nav-toggle-label + .nav-menu {
  display: block;
}

.nav-menu .navbar-nav {
  flex-direction: column !important;
  align-items: stretch;
  gap: 0.25rem !important;
  padding: 0.5rem 0;
}

.nav-menu .nav-item {
```

```
    width: 100%;
  }

  .nav-menu .nav-link {
    display: block;
    padding: 0.75rem 0.5rem;
    border-radius: 0.5rem;
  }
}
```

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All Products

8 models available



NEW M3 MAX

MacBook Pro 16"

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From $2,499

Add to Bag



THIN & POWERFUL

MacBook Air 15"

Incredibly thin design and up to 18 hours of battery life.

From $1,299

Add to Bag



ULTRA RETINA

iPad Pro

A stunningly advanced tablet for work and creativity.

From $999

Add to Bag



TITANIUM

iPhone 15 Pro

The most powerful iPhone with pro-level camera features.

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Add to Bag



ADVENTURE

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Add to Bag



NOISE CANCELING

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Immersive sound with Adaptive Audio and comfort fit.

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Add to Bag



WORKSPACE

Magic Keyboard

Comfortable typing and seamless connectivity for your setup.

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DISPLAY

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All Products



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Add to Bag



MacBook Air 15"

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Add to Bag



iPad Pro

From $999

Add to Bag



iPhone 15 Pro

From $999

Add to Bag



Apple Watch Ultra

From $799

Add to Bag



AirPods Pro

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Add to Bag



Magic Keyboard

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Add to Bag



Studio Display

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Store Mac iPad iPhone Watch AirPods Accessories 0

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All Products

8 models available



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THIN & POWERFUL

MacBook Air 15"

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ADVENTURE

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